TECHNICAL SPECIFICATION & MIGRATION GUIDE

Inverted MAML: PCBA v1.3.2 to 130nm ASIC Scaling

Patent Reference: SE 2630397-4 (Pat. Pend.) **Target Silicon:** SkyWater 130nm (SKY130) / Tiny Tapeout

Date: November 8, 2026

1. PHYSICAL ANALYSIS: PCBA STRESS TEST VS. SILICON REALITY

On the physical discrete testbed (PCBA v1.3.2), the 2T1C analog gain cells undergo an intentional forced discharge torture-test using C_W **= 100 nF** and R_{leak} **= 1 MΩ**. This was designed as a severe benchmark to induce measurable capacitor decay during bench evaluation.

PCBA v1.3.2 (Forced Torture Test):
$I_{leak,\ PCB}$ = 2.6 V / 1 MΩ = 2.6 µA
$(dV/dt)_{PCB}$ = 2.6 µA / 100 nF = **26.0 mV/ms** (corresponds to **6.65 LSB/ms** at 3.9 mV/LSB)

130nm SKY130 Silicon (Thick-Oxide 3.3V NMOS, No parallel resistor):
• At 25 °C (Room temperature, I_{leak} ≈ 30 fA): (dV/dt) = 30 fA / 200 fF = **0.15 mV/ms** (0.038 LSB/ms)
• At 85 °C (High die temp, I_{leak} ≈ 1.0 pA): (dV/dt) = 1.0 pA / 200 fF = **5.0 mV/ms** (1.28 LSB/ms)

Key Empirical Takeaway: At room temperature, the discrete PCBA leaks **170× faster** per millisecond than integrated 130nm silicon cells. Because online SGD with Forward Hadamard Propagation (FHP) stably converges and maintains high precision (peaking at 6.73 bits ENOB) over a 1.0 ms window on PCB, the algorithm is proven resilient. A window of **0.2–0.5 ms** on PCBA fully emulates a complete 10 ms refresh cycle in silicon.

2. SCALING SPECIFICATION: PCBA V1.3.2 VS. SKY130 ASIC

Parameter	PCBA v1.3.2 (Discrete)	130nm ASIC (SKY130 Target)	Physical Rationale & Technical Effect
Supply Voltage (V_{DD})	3.3 V	3.3 V (Analog IO) / 1.8 V (Digital)	Preserves identical analog operating headroom and dynamic range.
Virtual Ground (V_{ref})	1.65 V	1.65 V	Identical balanced midpoint for ColTIA and differential stages.
Weight Swing (V_W)	2.1 V – 3.1 V (1.0 V range)	1.8 V – 2.8 V (1.0 V range)	Full 1.0 V swing biased above threshold for stable conductance.
Row Input Swing (V_{in})	1.65 V – 1.90 V (250 mV)	1.65 V – 1.90 V (250 mV)	Keeps multiplier transistor deep in the linear triode region.
Storage Capacitor (C_W)	100 nF (SMD 0402 ceramic)	200 fF (MIM Cap M3/M4)	500,000× area reduction. Thermal noise √(kT/C) = 143 μV_{rms} (>11 bit SNR).
Perturbation Cap (C_{PTU})	1.0 nF – 2.2 nF	2.0 fF (Integrated MIM)	Preserves exact 1:100 capacitive division for non-destructive FHP.
Peak Column Current	5.0 mA	5.0 µA – 15 µA	1000× power reduction; matched for integrated CMOS summation.
ColTIA Feedback Resistor (R_F)	100 Ω (MCP6024T)	100 kΩ (High-sheet Poly)	Maintains identical 0.5 V output swing (5 µA × 100 kΩ = 0.5 V).

3. SELECTED PRIMITIVES IN SKYWATER 130NM PDK (`SKY130_FD_PR`)

- **Write Access FET (M_{write}):** `sky130_fd_pr__nfet_g5v0d10v5` with minimum width (W = 0.42 µm, L = 0.50 µm). Minimizes drain junction capacitance and restricts subthreshold leakage to < 50 fA at 25 °C. Coupled with a half-sized dummy FET to cancel clock feedthrough charge injection.
- **Multiplier Triode FET (M_{mul}):** `sky130_fd_pr__nfet_g5v0d10v5` with long channel (W = 0.50 µm, L = 4.0 µm). Long channel topology suppresses short-channel effects, optimizes Pelgrom matching, and sets the cell current to ~1.5–2.5 µA in deep triode.
- **MIM Capacitors:** `sky130_fd_pr__cap_mim_m3_1` (2.0 fF/µm²). C_W = 200 fF occupies 10 × 10 µm in M3/M4 metal layers directly above the FET channel, consuming zero base silicon area; C_{PTU} = 2 fF measures 1.0 × 1.0 µm.

- **Reference Discharge Path (Claim 8):** Replaced with a weakly biased MOSFET (W = 0.42 µm, L = 2.0 µm) operating in deep subthreshold ($V_{gate} \approx 0.45$ V), sinking a constant ~40 pA to sweep the full analog swing over 5 ms.

4. ON-CHIP MIXED-SIGNAL PINOUT FOR TINY TAPEOUT (8 ANALOG I/O)

By integrating the digital Hadamard generator and 6-bit R-2R ladders on-chip, along with an analog 2:1 CMOS multiplexer (transmission gates) for column readouts, the entire 6×6 triad fits within a single standard analog tile:

Analog Pins (8 Dedicated ua[7:0]):
• `ua[0..5]`: Analog column vector (muxed between ADC45 & ADC49)
• `ua[6]`: V_{prog} (Global analog programming line for 2T1C weights)
• `ua[7]`: V_{ref} / AGND (1.65 V virtual ground reference)

Digital Pins (Control from FPGA):
• `clk, rst_n, hadamard_en`: Synchronization and code trigger
• `spi_data, spi_clk, spi_latch`: Shifts digital inputs and weight addresses
• `adc_source_sel`: Selects Layer 1 (ADC45) or Layer 2 (ADC49)

5. PRE-ASIC SILICON VERIFICATION CHECKLIST ON PCBA V1.3.2

1. **Narrow Temporal Snapshot Window (0.4–0.5 ms):** Benchmark strata between 4.8 ms and 5.3 ms on PCBA v1.3.2. This directly mirrors > 2 ms continuous operation at 85 °C in silicon, providing definitive algorithmic validation.
2. **Time-Multiplexed ADC45 / ADC49 Readout:** Verify in FPGA firmware that online SGD converges identically when Layer 1 (ADC45) and Layer 2 (ADC49) are sampled across alternate cycles. This validates the single 6-channel external ADC architecture.
3. **Simulation in xschem / ngspice:** Test a single 2T1C cell with SKY130 BSIM4 models (W/L = 0.5/4.0, C_W = 200 fF) to confirm linear triode transconductance and < 2 µs transient settling.